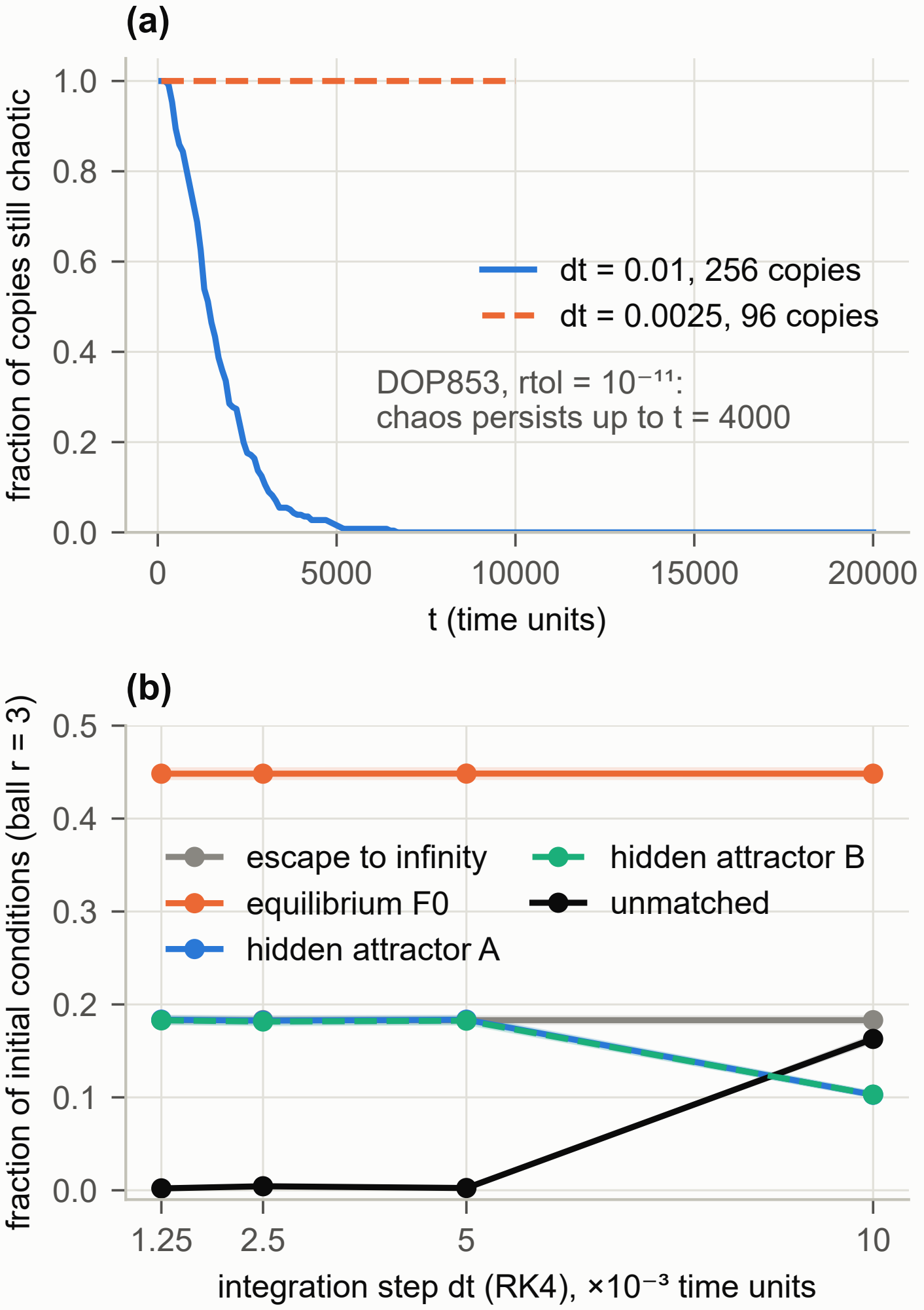
(a)
fraction of copies still chaotic
1.0
0.8
0.6
0.4
0.2
0.0
0
5000
10000
15000
20000
t (time units)
dt = 0.01, 256 copies
dt = 0.0025, 96 copies
DOP853, rtol = 10⁻¹¹:
chaos persists up to t = 4000
(b)
fraction of initial conditions (ball r = 3)
0.5
0.4
0.3
0.2
0.1
0.0
escape to infinity
equilibrium F0
hidden attractor A
hidden attractor B
unmatched
1.25
2.5
5
10
integration step dt (RK4), ×10⁻³ time units